# Do Post-Training Algorithms Actually Differ on Mathematical Reasoning at Scale?

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## Abstract

Practitioners routinely benchmark post-training algorithms at small scale, then deploy the winner at large scale. We frame the question as a *cross-scale study of post-training algorithms*—tracking how algorithm rankings evolve as a function of model size, without fitting a parametric scaling law—and show small-scale benchmarking is *anti-predictive* for mathematical reasoning: the within-family ordering of methods inverts as models grow. In a controlled study spanning $\sim$350 training runs across two model families (Qwen 2.5 0.5B–14B, Gemma 3 1B–12B), 8 algorithms, and 20 DPO variants with up to $N$=5 seeds, we find three results. **(1) Rankings invert with scale**: at $\leq$1.5B, Self-Play RFT (SP-RFT; standard rejection sampling fine-tuning (Yuan et al., 2023) on the model's own correct trajectories) is the strongest offline method; at 7B+, SP-RFT collapses to worst while DPO leads ($p < 0.001$). The inversion replicates in Gemma 3 and persists at Qwen 14B. **(2) Initialization dominates algorithm choice**: switching Instruct$\rightarrow$Base compresses the spread 3–15$\times$. **(3) Apparent differences have two regimes**: format compliance at small scale, architecture-specific reasoning at large scale. Together, these findings imply that small-scale algorithm selection is unreliable for mathematical reasoning, and practitioners should validate at deployment scale.

## 1 Introduction

Suppose you are choosing a post-training algorithm for mathematical reasoning. You benchmark Self-Play RFT (SP-RFT; rejection sampling fine-tuning (Yuan et al., 2023) on self-generated correct trajectories), DPO, and SimPO at 1.5B parameters: SP-RFT wins by 5 pp over DPO. You select SP-RFT, scale to 7B, and deploy—only to discover that SP-RFT is now the *worst* method by 6 pp, while DPO has silently become best. Your small-scale selection was not just uninformative; for the SFT-vs-preference axis it was anti-predictive—small-scale ranks invert at deployment scale (Figure 1).

**We frame this as a cross-scale study of post-training algorithms**: rather than chase a single best loss function, we measure how algorithm rankings evolve with model size and treat the curves themselves—not the point estimates—as the object of study. We use the term *scaling study* (not scaling *law*) deliberately: we do not fit a parametric form for $\mathrm{accuracy}(N)$, since the within-Qwen DPO–SP-RFT gap narrows from $+6.0$ pp at 7B to $+2.2$ pp at 14B—a non-monotone behavior that any single power/log fit would obscure.

This is not a hypothetical. It is the central empirical finding of this paper, replicated across two model families and confirmed with formal statistical testing ($p < 0.001$, $N$=5 seeds). The practitioner's dilemma is real: every published comparison operates at a single scale, yet deployment happens at a

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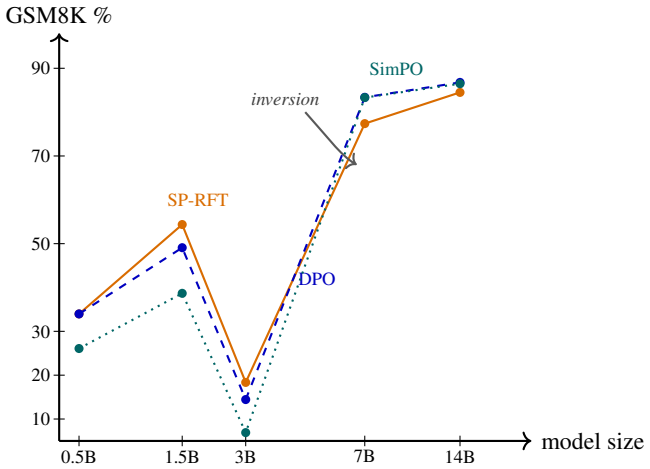


Figure 1: **Cross-scale ranking of post-training algorithms.** GSM8K strict-match accuracy vs. model size (log axis) for three offline post-training methods on Qwen 2.5, all at the shared default LR $10^{-6}$. The deep dip at 3B is a checkpoint-specific format-compliance pathology of Qwen 2.5-3B-Instruct (Appendix O); the substantive crossover is between 1.5B (SP-RFT leads) and 7B+ (DPO/SimPO lead). Small-scale rankings are not predictive of deployment-scale rankings; the inversion replicates on Gemma 3 (1B–12B; Section 4.2).

different one. If rankings invert, the entire methodology of algorithm benchmarking at small scale is called into question.

Post-training of large language models—adapting a pretrained model for tasks such as mathematical reasoning—has become the decisive stage separating a capable base model from a useful one (Ouyang et al., 2022; Touvron et al., 2023). Many algorithms compete for adoption: online RL methods like GRPO (Shao et al., 2024), offline preference methods like DPO (Rafailov et al., 2023), and a growing family of variants—SimPO (Meng et al., 2024), KTO (Ethayarajh et al., 2024), ORPO (Hong et al., 2024)—each claiming improvements over prior work. DeepSeek-R1 (DeepSeek-AI, 2025) demonstrated that pure RL post-training can elicit sophisticated reasoning, further expanding the design space. Yet each paper evaluates on different models, datasets, and scales, making cross-method comparison unreliable. The few existing studies (Xu et al., 2024; Spangher et al., 2025) cover limited methods at a single scale—they cannot detect the ranking inversions we observe.

We address this gap through a large-scale, controlled comparison using a unified framework where all algorithms share identical infrastructure. Our thesis is sharp: **on mathematical reasoning, post-training algorithm rankings invert with scale, making small-scale selection anti-predictive.** We emphasize that this is a task-specific phenomenon: on multiple-choice general benchmarks (ARC-C, HellaSwag, WinoGrande), algorithms differ by $<$1.2 pp at all scales (Section 4.4; Appendix T), though generative benchmarks remain untested. The evidence for mathematical reasoning unfolds in three findings:

1. **Rankings invert with scale—from Instruct checkpoints, across architectures.** On Qwen 2.5 Instruct, SP-RFT leads offline methods at $\leq$3B but collapses to worst at 7B, where DPO (83.4% $\pm$0.56) and SimPO (83.3% $\pm$1.79) dominate ($p < 0.001$ vs. SP-RFT, $N$=5). On Gemma 3 IT, the same pattern holds: SP-RFT $>$ DPO at 1B and 4B, but DPO $\geq$ SP-RFT at 12B (87.4% vs. 86.7%).
2. **Initialization explains most of the variance.** Replacing Instruct with Base initialization collapses the inter-algorithm spread from 15.7 pp to 1.07 pp at 1.5B and from 6.00 pp to 1.82 pp at 7B (a 3–15$\times$ compression). The dominant source of inter-algorithm variance is how each loss function interacts with prior alignment, not intrinsic properties of the objectives.
3. **Apparent differences have two regimes: format compliance at small scale, architecture-specific reasoning at large scale.** At 1.5B, the 19.3 pp GSM8K spread collapses 36$\times$ to 0.54 pp on MATH and 41$\times$ to 0.47 pp on general benchmarks—algorithms differ only in format compliance. At larger scales ($\geq$4B), MATH reveals genuine reasoning differences, but these depend on the

(algorithm, architecture) pair: DPO gains +9 pp on Gemma 12B MATH while SimPO gains +13 pp on Qwen 14B MATH. Both regimes yield the same practical conclusion: small-scale benchmarking is unreliable.

These findings are based on $\sim$323 training runs on Qwen 2.5 (0.5B–14B), supplemented by 27 runs on Gemma 3 (1B–12B), spanning 8 algorithms, 20 DPO variants, 4 evaluation domains (GSM8K, MATH, ARC-C/HellaSwag/WinoGrande), and up to $N{=}5$ seeds with formal statistical testing. The implication for practitioners is direct: validate at deployment scale, not prototype scale. The implication for the research community is that designing new loss functions at small scale—the dominant paradigm—may be selecting for the wrong properties.

## 2 Related Work

**Post-training algorithms and comparative studies.** The RLHF pipeline (Ouyang et al., 2022) uses PPO (Schulman et al., 2017); GRPO (Shao et al., 2024) simplifies this via group-normalized rewards. Wu et al. (2025) show a theoretical correspondence between GRPO and DPO; our empirical results show this correspondence is scale-conditioned (SGRPO leads DPO by +8.9 pp at 1.5B but trails by $-2.8$ pp at 7B), illustrating that scale-dependent empirical separations can coexist with their scale-free theoretical equivalence. DPO (Rafailov et al., 2023) bypasses reward modeling, spawning IPO (Azar et al., 2024), SimPO (Meng et al., 2024), KTO (Ethayarajh et al., 2024), ORPO (Hong et al., 2024), and others. We provide the first controlled evaluation of 20 DPO variants under identical conditions. Xu et al. (2024) and Ivison et al. (2024) cover only 2–3 algorithms. Spangher et al. (2025) benchmark 17 algorithms at a single scale—they cannot detect ranking inversions. Saeidi et al. (2025) evaluate DPO variants without multi-seed runs or multiple comparison correction. Our finding that variant rankings are scale-dependent is reminiscent of Lucic et al. (2018), who found that GAN variant rankings change across experimental conditions. Critically, none of these studies evaluate across multiple model families, leaving the question of architectural generality unanswered. Our Gemma 3 validation addresses this gap. Tan et al. (2025) study scaling behaviors of RL post-training; our work complements theirs by showing that *relative* rankings—not just absolute performance—are scale-dependent.

## 3 Methods and Experimental Setup

**Framework.** Our framework implements 51 post-training algorithms (the full list is enumerated in [anon repo]; this paper evaluates 28: 8 core methods plus 20 DPO variants, with the remaining 23 archived for future comparisons). All algorithms share identical model loading (HuggingFace, `flash_attention_2`), data pipelines, distributed training (DeepSpeed ZeRO-3 (Rajbhandari et al., 2020), vLLM (Kwon et al., 2023) for RL rollouts), and checkpointing. Each algorithm implements a `BaseAlgorithm` interface requiring only the loss function, ensuring the *only* difference between runs is the loss computation. We organize algorithms into: **(1) SP-RFT; (2) Online RL** (GRPO variants); **(3) Offline preference** (DPO (Rafailov et al., 2023), IPO (Azar et al., 2024), SimPO (Meng et al., 2024), KTO (Ethayarajh et al., 2024), and 16 additional variants); **(4) Hybrid** (ORPO (Hong et al., 2024), SPIN (Chen et al., 2024)). The 20-variant taxonomy is in Appendix A. **SGRPO** (Self-play GRPO) is our online-RL baseline: a token-level GRPO (Shao et al., 2024) variant that performs on-policy rollouts against the same self-play prompt set used by the offline methods, scoring each rollout with exact-match correctness and applying group-normalized advantages without an explicit KL term (an ablation in Appendix P shows KL has $<$0.2 pp effect). It serves as a controlled online counterpart to SP-RFT on identical data and prompts. Two further GRPO variants—**GSPO** (sequence-level importance ratio, robust on MoE-style routing noise) and **CISPO** (clipped ratio used as a detached weight on the log-probability)—are reported as ablations in Appendix E; main-table "online RL" numbers refer to SGRPO unless noted.

**Evaluation.** GSM8K (Cobbe et al., 2021) (1,319 test problems, exact-match, greedy decoding). MATH (Hendrycks et al., 2021) (5,000 problems, 4-shot) at 1.5B and general-domain benchmarks (ARC-C, HellaSwag, WinoGrande) at both 1.5B and 7B. All evaluations use the LM Evaluation Harness (Gao et al., 2024). We report means $\pm\sigma$ and perform Welch's $t$-tests with Bonferroni correction where indicated.

**Qwen 2.5 (primary).** Qwen 2.5 (Qwen Team, 2025) Instruct at 0.5B, 1.5B, 3B, 7B, and 14B. Our experiments comprise: (1) core comparison of 8 algorithms across 4 scales ($\sim$70 runs, $N$=5 seeds for core methods at 7B), with a full-FT/LoRA factorial at both 3B and 7B; (2) DPO variant taxonomy at 1.5B (20 variants $\times$ 5 seeds = 100 runs) and a stratified subset of 5 at 7B (15 runs); (3) 14B deployment-scale validation with SP-RFT, DPO, SimPO ($N$=3 seeds each); (4) ablations ($\sim$57 runs), including Base-vs-Instruct initialization and hyperparameter sensitivity; (5) a fixed-dataset control training 1.5B on 14B self-play data (6 runs, Section 4.5); (6) LR sweeps at 1.5B, 3B, and 7B (54 runs, Appendix M–O); (7) a gold-data SFT control using MetaMathQA at 1.5B and 7B (6 runs, Section 4.5). Total Qwen-side: $\sim$323 training runs, $\sim$640 GPU-hours on H100s (training only; the $\sim$940 GPU-hour grand total in Appendix I also includes evaluation). Together with the Gemma 3 runs below, the grand total is $\sim$350 training runs.

**Gemma 3 (cross-architecture validation).** To test whether our findings generalize beyond Qwen, we train SP-RFT, DPO, and SimPO on Gemma 3 (Gemma Team, 2025) IT at 1B, 4B, and 12B with 3 seeds each (27 runs). This uses the same GSM8K data and evaluation protocol; 12B uses LoRA, matching the Qwen 7B setup.

**Data.** All algorithms share the same data pipeline to eliminate data-source confounds. For each prompt in GSM8K's 7,473 training problems, we sample 16 responses from the *target model* at temperature 1.0 and score each with exact-match correctness. DPO/SimPO/IPO receive (chosen, rejected) pairs from the highest/lowest scoring responses. SP-RFT trains on the *chosen* trajectories from this same self-play dataset—not gold responses—so only the loss function differs between methods.[1] Because SP-RFT trains on model-generated chosen trajectories rather than gold or expert data, it is functionally *rejection sampling fine-tuning* (RFT) (Yuan et al., 2023); we use the SP-RFT label to distinguish it from standard SFT on gold data (Section 4.5). This distinction matters mechanistically: at small scale, where few model samples are correct, RFT on rare successes yields large gains; at large scale, where the model already solves most problems, RFT encounters a ceiling effect while DPO's contrastive signal can still extract improvements (see Section 5).

**Design rationale.** Data is generated *at each scale*, matching how practitioners train (Shao et al., 2024; DeepSeek-AI, 2025; Chen et al., 2024). Dataset quality therefore correlates with model scale; we address this confound with a fixed-dataset control (Section 4.5). **Training.** AdamW, cosine LR ($10^{-6}$) with 10% warmup, 3 epochs, batch size 8. LoRA (Hu et al., 2022) ($r=16$, $\alpha=32$) at $\geq$7B; full FT at smaller scales. BF16 on H100 80GB; $\beta=0.1$, SimPO $\gamma=0.5$. **Learning rate.** We sweep LR for SP-RFT and DPO at 1.5B, 3B, and 7B (Appendix M–O). The default $10^{-6}$ is optimal at 7B; higher rates degrade performance monotonically, with DPO more sensitive than SP-RFT (DPO collapses to 16% at $10^{-5}$ vs. SP-RFT at 53%). The ranking inversion (DPO $>$ SP-RFT) is preserved at independently optimized LRs for each algorithm. At 1.5B, SP-RFT leads at every LR tested, with DPO collapsing at $\geq 5 \times 10^{-6}$.

# 4 Results

## 4.1 Finding 1: Rankings Invert with Scale

Table 1 presents our main results on Qwen 2.5. At **0.5B**, five of six methods converge to identical strict-match accuracy across seeds ($\sigma$=0 for SP-RFT/DPO/KTO/SimPO/SGRPO; only IPO has $\sigma$=0.26). This is a *noise-floor* effect—0.5B is below the protocol's differentiation threshold—rather than algorithmic equivalence; we retain it for completeness but the substantive scaling signal begins at 1.5B (Appendix K). At **1.5B**, SGRPO (online RL) tops all methods at $58.0\% \pm 0.57$. Among offline methods, SP-RFT leads ($54.4\% \pm 0.59$), followed by IPO, KTO, DPO, and SimPO. At **3B**, the default LR ($10^{-6}$) produces catastrophic format non-compliance (base model: 62.9% flexible but 10.7% strict-match)—a checkpoint-specific pathology of Qwen 2.5-3B-Instruct that we acknowledge as a limitation. We report the default-LR numbers in Table 1 (consistent protocol) and a per-algorithm

[1] **Effective dataset overlap.** A prompt enters the SP-RFT set if at least one of its 16 samples is correct, and enters the DPO/SimPO/IPO set if it has both a correct and an incorrect sample. The two sets coincide on the dominant central regime where pass@16 is neither 0 nor 1; they diverge only in the tails (very low or very high base accuracy). At 0.5B/1.5B/7B Instruct on Qwen 2.5 the asymmetric drops are small ($< 5\%$ of prompts) and affect both methods directionally similarly, leaving the inversion intact; per-scale effective-dataset sizes will be released with the per-seed outputs.

Table 1: GSM8K exact-match accuracy (%) across scales (Qwen 2.5). 1.5B: 3-seed means $\pm\sigma$; 7B: 5-seed means for SP-RFT/DPO/SimPO, 3-seed for IPO/KTO. Best per scale in **bold**. [b]Single seed. [c]Epoch-2 checkpoint. [d]Best of 3 epochs. [a]62.9% flexible but 10.69% strict-match. **The 3B column reports the shared-default LR ($10^{-6}$) used at all other scales**; this matches the controlled-comparison protocol but is at the bottom of a hyperparameter pathology specific to the 3B Qwen 2.5 Instruct checkpoint. With per-algorithm best-LR tuning (Appendix O), SP-RFT reaches 55.7% at $10^{-5}$ and DPO 34.6% at $5\times10^{-6}$; SP-RFT>DPO at *every* LR tested. The default-LR numbers are reported here to keep Table 1 as a controlled comparison.

| **Algorithm** | **0.5B** | **1.5B** | **3B (full FT)** | **7B (LoRA)** |
|---|---|---|---|---|
| Base Model | 19.56 | 31.16 | 10.69[a] | 75.82 |
| SP-RFT | $33.97_{\pm0.0}$ (+14.4) | $54.36_{\pm0.59}$ (+23.2) | **18.35**[b] (+7.7) | $77.38_{\pm1.11}$ (+1.6) |
| DPO | $33.97_{\pm0.0}$ (+14.4) | $49.08_{\pm0.61}$ (+17.9) | $14.44_{\pm2.01}$ (+3.8) | **83.38**$_{\pm0.56}$ (+7.6) |
| IPO | **34.50**$_{\pm0.3}$ (+14.9) | $52.24_{\pm0.22}$ (+21.1) | — | $80.39_{\pm0.92}$ (+4.6) |
| KTO | $33.81_{\pm0.0}$ (+14.3) | $51.15_{\pm1.77}$ (+20.0) | — | $80.24_{\pm0.16}$ (+4.4) |
| SimPO | $26.08_{\pm0.0}$ (+6.5) | $38.67_{\pm1.78}$ (+7.5) | — | $83.32_{\pm1.79}$ (+7.5) |
| SGRPO | 32.45 (+12.9) | **58.00**$_{\pm0.57}$ (+26.8) | 8.11[c] (−2.6) | 80.59[d] (+4.8) |

best-LR sweep in Appendix O.[2] **SP-RFT leads at every LR tested**, preserving the small-scale ranking; the substantive claim is that SP-RFT>DPO at ≤1.5B and inverts at 7B+, with 3B as a noisy interior point.

The ordering *inverts* at **7B** (Figure 2). With $N$=5 seeds for SP-RFT, DPO, and SimPO, DPO reaches $83.38\% \pm 0.56$ and SimPO $83.32\% \pm 1.79$; a Welch's $t$-test fails to detect a significant difference ($p = 0.95$; note that a high $p$-value indicates failure to reject the null, not confirmation of equivalence—our sample size may lack power to resolve small gaps). The critical finding: SP-RFT drops from best offline method to *worst* at 7B ($77.38\%\pm1.11$), separated by ∼6 pp from DPO/SimPO ($p < 0.001$ after Bonferroni correction over the family of 10 pairwise tests at 7B; $\alpha = 0.005$, see Appendix K). This is the SP-RFT-to-preference inversion: the method that wins at small scale loses at large scale.

**Disentangling scale from LoRA.** The headline curve mixes full FT at ≤3B with LoRA at ≥7B—a practitioner-realistic choice (full FT is infeasible at 7B+ on common hardware) that confounds scale with training method. A 2×2 factorial at 7B trains SP-RFT/DPO/SimPO under *both* regimes: DPO leads under Full FT (84.5%) just as under LoRA (83.4%), and the inversion appears in both cells (Appendix D), so **scale**, not LoRA, drives the inversion; the mixed-method curve remains a residual limitation that full-FT runs at every scale could close.

### 4.2 Cross-Architecture Validation and Deployment Scale

**Gemma 3 (1B–12B).** A natural objection is that the ranking inversion may be a Qwen-specific artifact. To test generality, we train SP-RFT, DPO, and SimPO on Gemma 3 IT (Gemma Team, 2025) at 1B, 4B, and 12B (3 seeds each, 27 runs). The pattern mirrors Qwen. At **1B**, SP-RFT dominates (26.7%), DPO falls to 21.5%, and SimPO collapses to 4.1%. At **4B**, the same ordering holds with wider gaps: SP-RFT ($75.30 \pm 0.53$) ≫ DPO ($57.77 \pm 10.76$) ≫ SimPO ($40.03 \pm 8.31$); Welch's $t$ gives SP-RFT>DPO at $p$=0.11 (limited power at $N$=3 with high DPO variance) and SP-RFT>SimPO at $p$=0.018. At **12B**—the critical test—DPO (87.4%) overtakes SP-RFT (86.7%), with Welch's $t$ giving DPO>SP-RFT at $p$=0.036 ($N$=3, uncorrected); the 0.7 pp gap is small but the directional flip is clear and replicates the Qwen pattern. SimPO (75.1%) remains behind ($p$=0.048 vs. DPO).

Two differences from Qwen are worth noting. First, SimPO does not converge to DPO at the largest scale in Gemma, suggesting that the DPO/SimPO parity observed at Qwen 7B may be architecture-specific. Second, the inversion point appears later in Gemma (12B vs. 7B), which may reflect

[2]Best-LR results: SP-RFT reaches 55.7% at $10^{-5}$, DPO 34.6% at $5\times10^{-6}$, SimPO 6.9% (best = default). The default-LR ordering and the best-LR ordering both have SP-RFT>DPO; this checkpoint pathology is Qwen-specific and we expect a controlled 3B point would require either a Base-model 3B run or a different-family 3B checkpoint (e.g., Llama 3.2 3B), neither of which is in our budget.

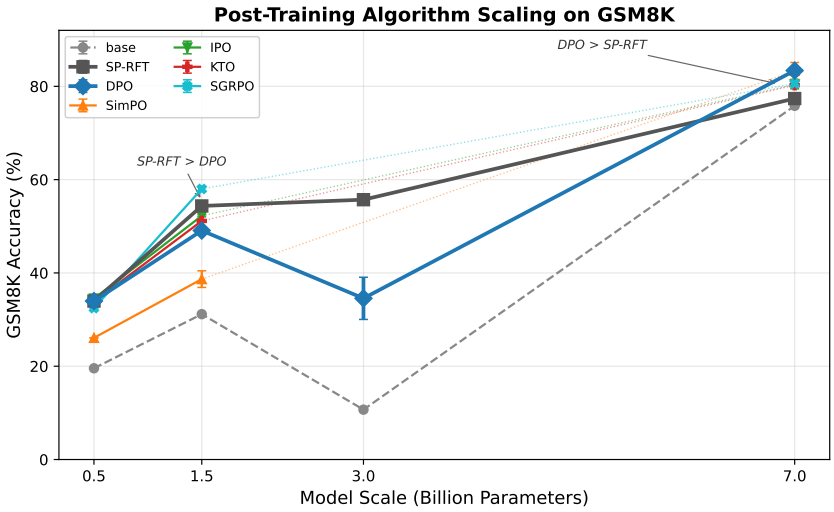


Figure 2: GSM8K accuracy across model scales (Qwen 2.5). At 7B, no significant difference is detected between DPO and SimPO ($p = 0.95$, $N$=5), while SP-RFT separates by ∼6 pp ($p < 0.001$). The SP-RFT-to-preference **ranking inversion** between ≤3B and 7B is the paper's central finding.

Table 2: Cross-architecture validation and deployment scale on GSM8K (%). Gemma 3: 3 seeds each; 12B uses LoRA. Qwen 14B: 3 seeds each, LoRA. The SP-RFT-to-DPO inversion replicates across architectures and persists at 14B.

| | **Gemma 1B** | | **Gemma 12B** | | **Qwen 14B** | |
|---|---|---|---|---|---|---|
| **Algorithm** | **Mean** | $\pm\sigma$ | **Mean** | $\pm\sigma$ | **Mean** | $\pm\sigma$ |
| SP-RFT | **26.7** | — | 86.7 | 0.29 | 84.5 | 0.19 |
| DPO | 21.5 | — | **87.4** | 0.26 | **86.8** | 0.23 |
| SimPO | 4.1 | — | 75.1 | 4.88 | 86.5 | 0.68 |
| *SP-RFT−DPO gap* | *+5.2 pp* | | *−0.7 pp* | | *−2.2 pp* | |

differences in the pretrained model's formatting capabilities. Despite these nuances, the central finding—that SP-RFT-to-preference rankings invert with scale—replicates across architectures.

**Qwen 14B.** To extend the scaling curve beyond 12B, we train SP-RFT, DPO, and SimPO on Qwen 2.5-14B-Instruct with 3 seeds each (9 runs). The inversion persists: DPO (86.8% ±0.23) > SimPO (86.5% ±0.68) > SP-RFT (84.5% ±0.19) > Base (79.5%). Welch's $t$-tests at $N$=3: DPO>SP-RFT significant ($p$=0.0002); SimPO>SP-RFT marginally significant ($p$=0.030); SimPO vs. DPO not detectable ($p$=0.53, gap 0.30 pp ≪ seed noise). Within the Qwen family the DPO–SP-RFT gap *narrows* from 6.0 pp at 7B to **2.2 pp** at 14B; on general benchmarks (ARC-C, HellaSwag, WinoGrande) the spread is <1.2 pp.

Notably, MATH evaluation at 14B reveals a different pattern: SimPO (39.1%) dramatically outperforms DPO (25.8%) and SP-RFT (23.0%), contrasting with Gemma 12B where DPO dominates MATH (50.2% vs. SimPO 32.7%). This architecture-dependent reversal on harder tasks further underscores that algorithm rankings cannot be extrapolated across models or tasks.

Table 2 presents the combined cross-architecture and deployment-scale results.

### 4.3 Finding 2: Initialization Dominates Algorithm Choice

Why do rankings invert? We hypothesize that algorithms interact differently with the prior alignment baked into Instruct checkpoints, and that removing this alignment should eliminate the differences.

Table 3 tests this at 1.5B. Starting from Base, all methods improve (SP-RFT: +6.4 pp, DPO: +12.2 pp, SimPO: +23.2 pp) and the spread collapses from 15.7 pp to just **1.07 pp**. SimPO—worst from

Table 3: Base vs. Instruct initialization at 1.5B on GSM8K (%). The spread collapses from 15.7 pp to 1.07 pp; SimPO reverses from worst to best. This compression generalizes to 7B (6.00 pp $\rightarrow$ 1.82 pp; Appendix Q). **Caveat:** Base-init values are single-seed point estimates; the 1.07 pp spread should therefore be read as suggestive (within plausible seed noise of $\sigma{\sim}0.6$ observed for 1.5B Instruct). The *direction* of the compression (Instruct$\gg$Base spread) is much larger than seed noise even on a generous reading.

| **Algorithm** | **From Base** | **From Instruct** | $\Delta$ |
|---|---|---|---|
| SP-RFT | 60.80 | $54.36_{\pm 0.59}$ | +6.4 |
| DPO | 61.33 | $49.08_{\pm 0.61}$ | +12.2 |
| SimPO | **61.87** | $38.67_{\pm 1.78}$ | +23.2 |
| *Spread* | *1.07 pp* | *15.7 pp* | |

Instruct—becomes best from Base, suggesting its reference-free nature makes it most sensitive to initialization.

Critically, this finding generalizes to 7B (Appendix Q): from Base, the spread collapses from 6.00 pp to 1.82 pp. The ranking from Base does *not* simply reproduce the 1.5B pattern: at 1.5B Base, SimPO leads (61.87%) followed by DPO (61.33%) and SP-RFT (60.80%); at 7B Base, SP-RFT leads (82.49%) followed by SimPO (80.82%) and DPO (80.67%). Thus, a residual ranking shift persists even from Base—but the key observation is that these differences are compressed to $\leq$1.82 pp, compared to 6–16 pp from Instruct. Initialization is the dominant source of inter-algorithm variance, even if it does not fully explain it.

DPO's sensitivity to Base initialization is theoretically expected (KL regularization anchors to a reference that cannot format math), but the convergence extends to reference-free methods (SimPO) and pure SP-RFT, so the prior alignment state—not just reference-model mechanics—is the primary source of inter-algorithm variance. Practically, most real-world post-training starts from Instruct, where algorithm choice does produce measurable differences; the Base ablations at both scales reveal that those differences are largely an initialization-loss interaction.

### 4.4 Finding 3: Two Regimes—Format Compliance, Then Architecture-Specific Reasoning

Tables 4 and 5 compare GSM8K and MATH results. At 1.5B, the 19.3 pp GSM8K spread collapses to **0.54 pp** on MATH—a $36\times$ compression. Rankings shift: SGRPO drops from 1st to 4th, SimPO rises from last to 2nd. On general-domain benchmarks (ARC-C, HellaSwag, WinoGrande), the spread collapses further to **0.47 pp** ($41\times$; Appendix G). At small scale, algorithms differ only in format compliance, not reasoning.

**But this picture changes at scale.** Table 5 shows MATH accuracy across Gemma 3 scales. At 1B, the raw spread is 15.5 pp, but this is driven entirely by SimPO's catastrophic collapse to 1.1%; excluding SimPO, Base/SP-RFT/DPO cluster within 2.9 pp. At 4B, DPO pulls ahead by +8.1 pp over SP-RFT. At 12B, DPO reaches 50.2%—nearly **+9 pp over base and +17 pp over SimPO**. The spread among non-pathological methods *increases* with scale, the opposite of the compression observed at 1.5B.

This pattern is *architecture-dependent*: at Qwen 14B, **SimPO** dominates MATH ($39.1\% \pm 2.2$) over DPO ($25.8\% \pm 0.4$) by $+13$ pp—the reverse of Gemma. Meanwhile general benchmarks (ARC-C, HellaSwag, WinoGrande; Appendix T) show $<$2 pp spread at all scales/architectures, but those are multiple-choice and do not test open-ended generation, so they do not rule out inversions on generative tasks (AlpacaEval, MT-Bench). Two-regime picture: at small scale algorithms differ shallowly (format); at large scale they differ deeply (reasoning), but *which algorithm wins depends on architecture and task*—both regimes make small-scale extrapolation unreliable.

### 4.5 Controls: Disentangling Data Quality and RFT Artifacts

**Fixed-dataset control.** To disentangle data quality from model scale, we train 1.5B Qwen 2.5 Instruct on self-play data generated by the 14B model. **SP-RFT still beats DPO by +5.1 pp** ($58.38\% \pm 0.49$ vs. $53.32\% \pm 0.70$, 3 seeds each); both algorithms gain comparably from higher-quality data (SP-RFT:

Table 4: Cross-task comparison at 1.5B: GSM8K vs. MATH. The 19.3 pp spread collapses to 0.54 pp (36× compression).

| Algorithm | MATH | GSM8K |
|---|---|---|
| SP-RFT | **27.04** | 54.36 |
| SimPO | 26.72 | 38.67 |
| KTO | 26.64 | 51.15 |
| SGRPO | 26.58 | **58.00** |
| DPO | 26.50 | 49.08 |
| *Spread* | *0.54* | *19.33* |

Table 5: MATH accuracy (%) across scales. At ≥4B, DPO leads in Gemma but SimPO leads in Qwen 14B.

| | Gemma 3 | | | Qwen |
|---|---|---|---|---|
| | 1B | 4B | 12B | 14B |
| Base | 16.6 | 34.4 | 41.3 | 16.8 |
| SP-RFT | 13.7 | 34.3 | 41.0 | 23.0 |
| DPO | 16.4 | **42.4** | **50.2** | 25.8 |
| SimPO | 1.1 | 39.2 | 32.7 | **39.1** |

Table 6: Fixed-dataset control: 1.5B on *14B* self-play data (GSM8K %). SP-RFT still leads by +5.1 pp.

| Algorithm | Mean | $\sigma$ |
|---|---|---|
| SP-RFT | **58.38** | 0.49 |
| DPO | 53.32 | 0.70 |
| *Gap* | *+5.06 pp* | |

Table 7: Gold-SFT control: standard SFT on MetaMathQA (GSM8K %). SFT advantage collapses from +12 pp to ≈0 pp with scale.

| | 1.5B | | 7B | |
|---|---|---|---|---|
| Method | Mean | $\sigma$ | Mean | $\sigma$ |
| Gold-SFT | **61.05** | 0.45 | 83.32 | 0.28 |
| SP-RFT | 54.36 | 0.59 | 77.38 | 1.11 |
| DPO | 49.08 | 0.61 | **83.38** | 0.56 |

+4.0 pp, DPO: +4.2 pp), leaving the gap stable (5.06 vs. 5.28 pp with own data). Model scale, not data quality, is the primary driver.

**Gold-data SFT control.** To rule out an RFT-ceiling artifact, we train standard SFT on MetaMathQA (Yu et al., 2024) (240K external expert solutions; 3 seeds at both 1.5B and 7B). Tables 6–7 report both controls side by side: at **1.5B**, Gold-SFT leads at $61.05\% \pm 0.45$ (+12.0 pp over DPO); at **7B**, Gold-SFT converges to DPO ($83.32\% \pm 0.28$ vs. $83.38\% \pm 0.56$). The SFT advantage collapses from +12 pp to ≈0 pp with scale regardless of whether the training data is self-generated or external—**falsifying the RFT-artifact hypothesis**.

### 4.6 Algorithm Selection Under Scale Shift

Given that small-scale rankings are anti-predictive, what should a practitioner do when target-scale validation is too expensive? Over the cross-scale ranks in Tables 1–2 we report two summary criteria: *worst-case rank* $R_{\text{wc}}{=}\max_N r_a^{(N)}$ and *deployment rank* $R_{\text{dep}}{=}\text{mean}_{N\geq 7\text{B}}\, r_a^{(N)}$. Values: SP-RFT ($R_{\text{wc}}{=}4, R_{\text{dep}}{=}3.5$), DPO $(4, 1)$, SimPO $(5, 2)$. DPO Pareto-dominates SP-RFT on $R_{\text{dep}}$ at equal $R_{\text{wc}}$, and dominates SimPO on both; Gemma 3 replicates the order. **DPO is therefore the rank-stable bet**—best from 7B onward across both architectures, lowest seed variance ($\sigma{=}0.56$ at 7B vs. SimPO's 1.79). Caveat: at 14B DPO ($86.78{\pm}0.23$) and SimPO ($86.48{\pm}0.68$) overlap within seed noise, so $R_{\text{dep}}$ should be read with bootstrap rank intervals (Appendix G) rather than as a hard ordering. Empirically the DPO–SP-RFT gap turns positive past an architecture-specific crossover (Qwen between 3B and 7B; Gemma between 4B and 12B), so use DPO when deployment scale is within 2× of that crossover; we avoid a parametric $\Delta(N)$ form because the within-Qwen gap *narrows* from +6.0 pp at 7B to +2.2 pp at 14B. The mechanistic basis (Section 5) is that SP-RFT hits a ceiling once base accuracy is high, while contrastive objectives still extract gradient from incorrect trajectories.

### 4.7 Practitioner Recommendations

Actionable guidance: **(1) Validate at deployment scale, architecture, and task**—rankings at ≤3B do not predict 7B+ behavior, and MATH rankings flip between Gemma 12B (DPO) and Qwen 14B (SimPO). **(2) DPO is a strong default on GSM8K**—near top at 1.5B/7B Qwen, overtakes SP-RFT at 12B Gemma and 14B Qwen, lowest seed variance ($\sigma{=}0.56$ at 7B), but does not transfer uniformly to MATH. **(3) Distrust variant sweeps at small scale**—the 1.5B→7B variant ranking fully inverts;

prefer vanilla DPO unless you can validate at target scale. **(4) Budget for multiple seeds**—SimPO matches DPO in mean at 7B but with $3.2\times$ the variance.

# 5 Discussion

**Why rankings invert: a mechanistic sketch.** The Base-vs-Instruct ablation (Section 4.3) identifies one mechanism: at small scale the Instruct model rarely produces correct solutions, so RFT on the few successful trajectories yields large gains while preference methods waste capacity modeling pairs of incorrect outputs; at 7B the Instruct model already solves most GSM8K problems and RFT hits a ceiling, while DPO's contrastive signal still extracts gradient from incorrect trajectories. The gold-data SFT control (Section 4.5) shows even standard SFT on MetaMathQA converges to DPO at 7B ($83.32$ vs. $83.38$) despite dominating at 1.5B by $+12$ pp, so the inversion is not merely a self-play data-saturation artifact but a genuine property of how supervised vs. contrastive objectives interact with model scale.

**What drives accuracy in mathematical reasoning.** In decreasing order of leverage: model scale ($\sim$50 pp), initialization ($\sim$15 pp), training paradigm (SP-RFT vs. preference: $\sim$6 pp at 7B), and loss function choice within the preference family ($<$1 pp for DPO vs. SimPO on GSM8K; up to 5.8 pp across DPO variants at 7B). On harder reasoning tasks (MATH), a new dimension emerges: algorithm-architecture interaction. DPO gains +9 pp on Gemma 12B MATH while SimPO gains +13 pp on Qwen 14B MATH—a complete reversal. This suggests that loss functions interact with architecture-specific representations in ways that are invisible on format-sensitive tasks like GSM8K.

**Methodological caveats at 7B+.** Four design choices interact: the fixed LR $10^{-6}$ is below the standard LoRA recommendation but an ablation rules out under-training (Appendix M); SGRPO was trained without explicit KL but adding $\lambda_{\mathrm{KL}}$=0.01 yields $+0.15$ pp (Appendix P); the 7B full-FT factorial (Appendix D) confirms the inversion under both regimes; and the effective batch size of 8 prompts is conservative for DPO, which already wins at 7B regardless. Online RL ablations (SGRPO, GSPO, CISPO) are in Appendix E.

# 6 Conclusion

On mathematical reasoning, post-training algorithm rankings invert with scale: the method that wins at 1.5B loses at 7B, replicated across Qwen 2.5 (up to 14B) and Gemma 3 (up to 12B), confirmed with $p < 0.001$ at $N$=5, and surviving fixed-dataset and gold-data SFT controls. On MC general benchmarks the spread is $<$1.2 pp at all scales—the inversion is task-specific. Initialization, not loss function, explains most algorithmic differences (3–15$\times$ spread compression from Base). The dominant paradigm of small-scale benchmarking + extrapolation is therefore unreliable for mathematical reasoning; practitioners should validate at deployment scale, and the community should invest in how algorithms interact with scale, initialization, and task structure rather than designing new losses at a single small scale.

**Limitations.** Two families (Qwen 2.5 14B, Gemma 3 12B) on math; the MC-only general-benchmark control limits transfer to generative tasks. The fixed-dataset control is asymmetric (no converse 7B-on-1.5B-data run). 0.5B is at the protocol's noise floor (Appendix K); 3B Qwen Instruct has a known LR pathology (Appendix O). The 20-variant DPO sweep uses shared $\beta$=0.1, LR= $10^{-6}$. Code, configs, per-seed JSONs release upon acceptance.

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# A Full DPO Variant Results

Table 8: Taxonomy of 20 DPO variants evaluated in this study, organized by primary modification to the DPO objective. All evaluated at 1.5B on GSM8K with 5 seeds.

| Variant | Primary Modification | Ref. | Year |
|---|---|---|---|
| DPO | Baseline logistic loss | Y | 2023 |
| IPO | Squared loss regularization | Y | 2024 |
| SimPO | Length-normalized, no ref | N | 2024 |
| KTO | Unpaired binary feedback | Y | 2024 |
| Hinge | Hinge loss replacement | Y | 2024 |
| CPO | Contrastive preference | N | 2024 |
| ORPO | SFT-integrated preference | N | 2024 |
| RDPO | Robust divergence penalty | Y | 2024 |
| CDPO | Conservative DPO | Y | 2024 |
| BetaDPO | Adaptive $\beta$ scheduling | Y | 2024 |
| CalDPO | Calibrated reward margin | Y | 2024 |
| DPOP | Pref.-weighted optimization | Y | 2024 |
| ODPO | Online DPO | Y | 2024 |
| EXO | Exponentiated gradient | Y | 2024 |
| AlphaPO | Alpha-divergence pref. | N | 2024 |
| APO | Anchored preference | Y | 2024 |
| SPPO | Self-play preference | Y | 2024 |
| RobustDPO | Dist. robust DPO | Y | 2024 |
| GPO | Generalized pref. opt. | Y | 2024 |
| FocalPO | Focal loss preference | Y | 2024 |

**Statistical notes.** Significance threshold: Bonferroni-corrected $\alpha = 0.05/19 = 0.0026$. Only SimPO ($p < 10^{-4}$) passes, and it is significantly *worse*. ORPO ($p = 0.013$) and DPOP ($p = 0.019$) would be significant uncorrected but fail after correction. Our 5-seed design provides 80% power to detect effects $\geq 1.0$ pp (Cohen's $d \geq 0.8$).

**At 7B: point-estimate rankings reverse.** We evaluate a stratified subset of 5 variants at 7B (3 seeds each), selected to span the 1.5B ranking (1st, 2nd, 7th, 13th, 15th). ORPO—1st at 1.5B—collapses to worst at 7B ($-4.79$ pp vs. DPO). EXO—15th at 1.5B—rises to the top ($+1.00$ pp). No variant reaches Bonferroni-corrected significance at either scale. The practical conclusion: since variant rankings are statistically unresolvable at both scales and their point estimates reverse, small-scale variant sweeps provide no reliable signal for large-scale behavior.

**Shared-hyperparameter design.** All 20 variants use the same LR ($10^{-6}$) and $\beta = 0.1$. This is a deliberate choice: the 20-variant sweep evaluates *default-hyperparameter transferability*—whether

Table 9: Full DPO variant results at 1.5B scale on GSM8K (5 seeds). $\Delta$ is the accuracy difference from vanilla DPO. $^\dagger$ indicates statistical significance after Bonferroni correction ($p < 0.0026$). Cohen's $d$ measures practical effect size.

| Variant | Category | Mean (%) | Std | $\Delta$ DPO | $p$ | $d$ |
|---|---|---|---|---|---|---|
| ORPO | Reference-free | 53.89 | 0.60 | +4.12 | 0.0134 | 2.48 |
| DPOP | Weighting/calibration | 53.62 | 1.80 | +3.85 | 0.0189 | 1.88 |
| SPPO | Data augmentation | 52.60 | 1.45 | +2.84 | 0.0521 | 1.49 |
| IPO | Alternative divergence | 52.36 | 0.75 | +2.59 | 0.0614 | 1.53 |
| CalDPO | Weighting/calibration | 52.16 | 1.58 | +2.40 | 0.0934 | 1.22 |
| CPO | Reference-free | 51.43 | 1.57 | +1.67 | 0.2187 | 0.85 |
| GPO | Alternative divergence | 50.98 | 2.41 | +1.21 | 0.4370 | 0.52 |
| RobustDPO | Robustness | 50.89 | 2.13 | +1.12 | 0.4442 | 0.51 |
| CDPO | Robustness | 50.72 | 1.89 | +0.96 | 0.4916 | 0.46 |
| KTO | Unpaired | 50.48 | 2.11 | +0.71 | 0.6211 | 0.33 |
| Hinge | Alternative divergence | 49.87 | 2.43 | +0.11 | 0.9449 | 0.05 |
| DPO | Vanilla | 49.76 | 2.27 | – | – | – |
| ODPO | Data augmentation | 48.37 | 3.20 | −1.39 | 0.4525 | −0.50 |
| APO | Data augmentation | 48.07 | 2.73 | −1.70 | 0.3175 | −0.68 |
| EXO | Data augmentation | 47.52 | 2.92 | −2.24 | 0.2145 | −0.86 |
| FocalPO | Weighting/calibration | 47.04 | 3.27 | −2.73 | 0.1686 | −0.97 |
| RDPO | Alternative divergence | 46.79 | 2.47 | −2.97 | 0.0835 | −1.25 |
| BetaDPO | Weighting/calibration | 45.75 | 4.31 | −4.02 | 0.1145 | −1.17 |
| AlphaPO | Reference-free | 41.68 | 5.13 | −8.08 | 0.0204 | −2.04 |
| SimPO$^\dagger$ | Reference-free | 38.23 | 2.12 | −11.54 | 0.0000 | −5.25 |

variants provide gains "out of the box" with shared settings—not absolute algorithmic capability ceilings. This matches how practitioners typically deploy variants: using the original DPO paper's defaults. Per-variant tuning (e.g., optimizing $\beta$ for IPO's squared loss or KTO's unpaired feedback) could reveal differences that our shared-hyperparameter design masks. Our conclusion (variants are unresolvable) applies to the plug-and-play setting; per-variant tuning could surface real differences that shared hyperparameters mask.

Table 10: Stratified DPO variant evaluation at 7B (3 seeds each). Variants were selected to span the 1.5B ranking (1st, 2nd, 7th, 13th, 15th) to test rank-order predictiveness. Vanilla DPO: 83.38% ($N$=5). Point-estimate rankings reverse: the 1.5B leader (ORPO) is the 7B laggard; the 1.5B underperformer (EXO) rises to top. No variant reaches Bonferroni-corrected significance at either scale.

| Variant | Mean (%) | Std | $\Delta$ DPO | $p$ | 1.5B Rank |
|---|---|---|---|---|---|
| EXO | **84.38** | 0.96 | +1.00 | 0.277 | 15th |
| ODPO | 83.88 | 1.31 | +0.49 | 0.654 | 13th |
| GPO | 83.40 | 1.05 | +0.02 | 0.984 | 7th |
| Vanilla DPO | 83.38 | 0.56 | — | — | 12th |
| DPOP | 80.92 | 1.18 | −2.46 | 0.085 | 2nd |
| ORPO | 78.59 | 1.04 | −4.79 | 0.013 | 1st |

# B Training Configurations

For models at 7B/12B and for the 3B factorial, we apply LoRA (Hu et al., 2022) with rank $r = 16$, $\alpha = 32$, applied to all linear layers. The core comparison at $\leq$3B/4B uses full fine-tuning.

# C Framework Engineering

Scaling online RL to 7B required resolving three critical infrastructure issues: **(1) Ray GPU over-subscription.** Fractional `num_gpus` values in Ray caused GPU over-subscription deadlocks during multi-worker rollout generation; we enforce integer GPU allocation per worker. **(2) vLLM memory**

Table 11: Shared training configuration across all experiments.

| Parameter | Value |
|---|---|
| Optimizer | AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$, $\epsilon = 10^{-8}$) |
| Weight decay | 0.01 |
| LR schedule | Cosine with 10% linear warmup |
| Effective batch size | 8 prompts ($1 \times 8$ grad. accum.) |
| Training epochs | 3 |
| Precision | BF16 mixed precision |
| Distributed strategy | DeepSpeed ZeRO Stage 3 |
| Hardware | $8\times$ NVIDIA H100 80GB HBM3 |
| Sequence length | 512 (math), 1024 (code) |

Table 12: Algorithm-specific hyperparameters.

| Algorithm | Learning Rate | Key Hyperparameters | GPUs/Run |
|---|---|---|---|
| SP-RFT | $1\times10^{-6}$ | – | 1 |
| DPO | $1\times10^{-6}$ | $\beta = 0.1$ | 1 |
| SimPO | $1\times10^{-6}$ | $\beta = 0.1$, $\gamma = 0.5$ | 1 |
| IPO | $1\times10^{-6}$ | $\beta = 0.1$ | 1 |
| KTO | $1\times10^{-6}$ | $\beta = 0.1$ | 1 |
| SGRPO | $1\times10^{-6}$ | $\lambda_{\text{KL}} = 0^*$, $n = 4$ | 2 |
| GSPO | $1\times10^{-6}$ | $\lambda_{\text{KL}} = 0$, $n = 4$ | 2 |
| CISPO | $1\times10^{-6}$ | $\lambda_{\text{KL}} = 0$, $n = 4$, clip $= [0.8, 1.2]$ | 2 |

*Following the default GRPO recipe. See Appendix P for an ablation with $\lambda_{\text{KL}} = 0.01$.

**leak.** Rollout output objects were not explicitly freed between batches, causing a slow memory leak that triggered OOM after ∼200 steps at 7B; explicit deletion with `gc.collect()` resolved this. **(3) Epoch-transition OOM.** Optimizer state accumulated across epoch boundaries; offloading optimizer state to CPU between epochs eliminated the failure.

# D LoRA Factorial

Table 13: 2×2 factorial disentangling scale from LoRA (Qwen 2.5). The inversion appears under both Full FT and LoRA at 7B: DPO (84.5%) > SimPO (83.6%) > SP-RFT (79.8%) under Full FT, mirroring the LoRA ranking. 7B LoRA values: 5-seed means $\pm\sigma$; Full FT: single seed.

| | | 3B | 7B |
|---|---|---|---|
| **Full FT** | Base | 10.69 | 75.82 |
| | SP-RFT | **18.35**$^b$ (+7.7) | 79.83 (+4.0) |
| | DPO | $14.44_{\pm 2.01}$ (+3.8) | **84.46** (+8.6) |
| | SimPO | $6.90^b$ (−3.8) | 83.62 (+7.8) |
| **LoRA** | Base | 10.69 | 75.82 |
| | SP-RFT | 18.04 (+7.4) | $77.38_{\pm 1.11}$ (+1.6) |
| | DPO | 16.00 (+5.3) | $\mathbf{83.38}_{\pm 0.56}$ (+7.6) |
| | SimPO | 6.90 (−3.8) | $83.32_{\pm 1.79}$ (+7.5) |

# E Online RL Explorations

SGRPO (online RL) leads at 1.5B ($58.0\% \pm 0.57$, +3.6 pp over SP-RFT) at ∼6× the compute cost. At 7B, SGRPO achieves 80.59% (best of 3 epochs), placing it above SP-RFT but below DPO/SimPO. Adding KL regularization ($\lambda_{\text{KL}} = 0.01$) yields only +0.15 pp (Appendix P), confirming that the online–offline gap is genuine rather than a regularization artifact. **Caveat:** SGRPO uses the same effective batch size (8 prompts) and fixed 3-epoch schedule as offline methods; online RL typically benefits from larger batches and dynamic rollout budgets, so our SGRPO results may underestimate

online RL's true ceiling. SGRPO was not evaluated at 12B or 14B due to its $\sim 6\times$ compute cost. Token-level SGRPO outperforms sentence-level GSPO by 12–18 pp; GSPO and CISPO results are below.[3]

**SGRPO at 3B.** SGRPO at 3B achieves 4.93% after epoch 1 and 8.11% after epoch 2, both below the 10.69% baseline. On-policy rollouts compound the format compliance crisis: poorly formatted generations receive no reward, preventing the policy from learning formatting.

**SGRPO at 7B.** SGRPO achieves 80.59% (best of 3 epochs; 80.59/80.29/79.91), above SP-RFT but below DPO/SimPO. Slight accuracy degradation at epochs 2–3 suggests overfitting to the reward signal.

**GSPO and CISPO — caveat: implementation status not externally validated.** GSPO (sentence-level importance ratio) achieves 39.58% at 1.5B and 19.86% at 0.5B—12–18 pp below token-level SGRPO. CISPO (clipped ratio as a detached weight, clip $[0.8, 1.2]$) fails catastrophically (0.15% at 0.5B, 5.23% at 1.5B). **We do not yet have an externally-published reference number for either GSPO or CISPO on Qwen 2.5 with our exact training recipe, so we cannot rule out implementation issues**—the most likely candidate is rollout/trainer log-probability drift in the sequence-level estimator (which GSPO is designed to absorb but CISPO is more sensitive to). We therefore report these numbers as observed under our framework but do not draw method-level conclusions from them; the main paper's claims rely on the offline methods and SGRPO. Re-validating GSPO/CISPO against an external benchmark is on our open-question list.

# F SimPO Variance Analysis

SimPO exhibits the highest seed variance of any method at any scale ($\sigma = 1.79$ at 7B vs. DPO's $\sigma = 0.56$). Across 5 seeds, SimPO's per-seed accuracy ranges from 81.50% to 85.82% (range: 4.32 pp), while DPO ranges from 82.79% to 84.23% (range: 1.44 pp)—a $3\times$ difference. SimPO's two highest and two lowest seeds cluster in pairs, consistent with a bifurcation in the optimization landscape. This bimodal pattern is absent in DPO. SimPO computes length-normalized log-probabilities without a reference model, lacking the implicit regularization of DPO's reference-anchored gradient. The practical consequence: SimPO's "best seed" appears to outperform DPO (85.82 vs. 84.23), but no significant difference is detected under proper multi-seed evaluation ($p = 0.95$).

# G Additional Figures and Tables

Table 14: General-domain evaluation at 1.5B: ARC-Challenge (acc_norm), HellaSwag (acc_norm), WinoGrande (acc). The 19.3 pp GSM8K spread collapses to 0.47 pp ($41\times$ compression).

| **Algorithm** | **ARC-C (%)** | **HellaSwag (%)** | **WinoGrande (%)** | **Avg (%)** |
|---|---|---|---|---|
| Base Model | 46.76 | 68.23 | 62.90 | 59.30 |
| SP-RFT | 46.16 | 68.16 | 62.90 | 59.08 |
| DPO | 46.67 | 68.38 | 62.67 | 59.24 |
| IPO | 46.76 | 68.36 | 62.67 | 59.26 |
| KTO | **47.01** | 68.25 | 63.06 | 59.44 |
| SimPO | 46.08 | **68.37** | 62.59 | 59.01 |
| SGRPO | 46.67 | 68.23 | **63.54** | **59.48** |
| *Spread* | *0.93* | *0.22* | *0.95* | *0.47* |

# H Format Compliance Analysis

Table 16 reports strict-match and flexible-extract accuracy. The *format compliance gap* (strict $-$ flexible) quantifies whether a model produces well-formatted answers.

[3] GSPO achieves 39.58% at 1.5B; CISPO fails catastrophically (5.23% at 1.5B, 0.15% at 0.5B).

Table 15: General-domain evaluation at 7B (LoRA). Despite a 6.0 pp GSM8K spread, the general-domain spread is just 0.71 pp.

| **Algorithm** | **ARC-C (%)** | **HellaSwag (%)** | **WinoGrande (%)** | **Avg (%)** |
|---|---|---|---|---|
| Base Model | 55.20 | 80.48 | 71.11 | 68.93 |
| SP-RFT | 55.29 (+0.09) | 80.45 (−0.03) | 70.72 (−0.39) | 68.82 (−0.11) |
| DPO | 54.86 (−0.34) | **80.58** (+0.10) | 70.72 (−0.39) | 68.72 (−0.21) |
| IPO | **55.55** (+0.35) | 80.37 (−0.11) | **71.35** (+0.24) | **69.09** (+0.16) |
| KTO | 55.03 (−0.17) | 80.39 (−0.09) | 70.72 (−0.39) | 68.71 (−0.22) |
| SimPO | 54.44 (−0.77) | 80.53 (+0.05) | 70.17 (−0.95) | 68.38 (−0.56) |
| *Spread (trained)* | *1.11 pp* | *0.21 pp* | *1.18 pp* | *0.71 pp* |

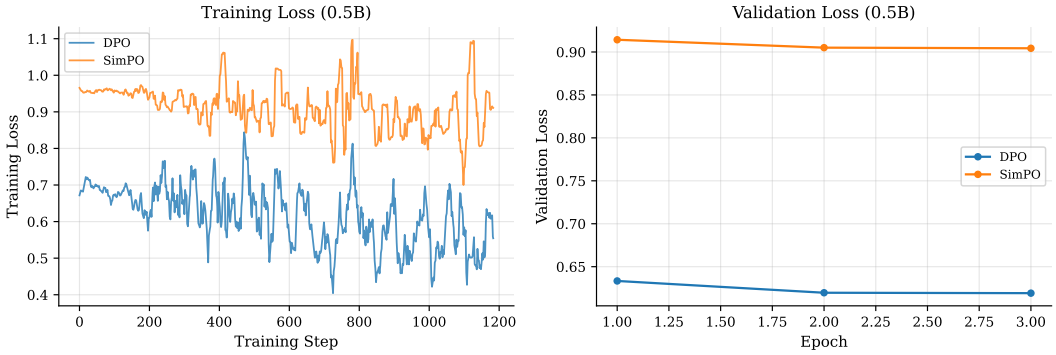


Figure 3: Training dynamics comparison at 0.5B scale. **Left:** training loss over steps (smoothed). **Right:** validation loss over epochs.

# I Compute Budget

The total compute (∼940 GPU-hours) is modest given the breadth (∼350 training runs across 2 model families, 8 scales, 4 evaluation domains) and substantially less than training a single 7B model from scratch.

# J Additional Scaling Results

SGRPO at 1.5B achieves $58.00\% \pm 0.57$ (seeds: 57.47%, 57.92%, 58.61%). SL methods at 1.5B use multi-seed reruns with genuine variance (Table 18).

# K Multi-Seed Verification

**Statistical tests at 7B.** Welch's $t$-tests with Bonferroni correction ($\alpha = 0.05/10 = 0.005$):

- **DPO vs. SimPO:** $\Delta = +0.06$ pp, $p = 0.950$. Not significant; no difference detected at this sample size.
- **DPO vs. SP-RFT:** $\Delta = +6.01$ pp, $p < 0.001$. **Significant.**
- **SimPO vs. SP-RFT:** $\Delta = +5.94$ pp, $p < 0.001$. **Significant.**
- **DPO vs. KTO:** $\Delta = +3.15$ pp, $p < 0.001$. **Significant.**
- **DPO vs. IPO:** $\Delta = +2.99$ pp, $p = 0.029$. Not significant after correction.
- **SP-RFT vs. IPO:** $\Delta = -3.01$ pp, $p = 0.018$. Not significant after correction.
- **SP-RFT vs. KTO:** $\Delta = -2.86$ pp, $p = 0.006$. Not significant after correction (marginal).
- **IPO vs. KTO:** $\Delta = +0.15$ pp, $p = 0.837$. Not significant.

Three groups emerge: (1) DPO/SimPO cluster (∼83.4%), (2) IPO/KTO cluster (∼80.3%), (3) SP-RFT (∼77.4%). The DPO/SimPO > SP-RFT separation is robust ($p < 0.001$).

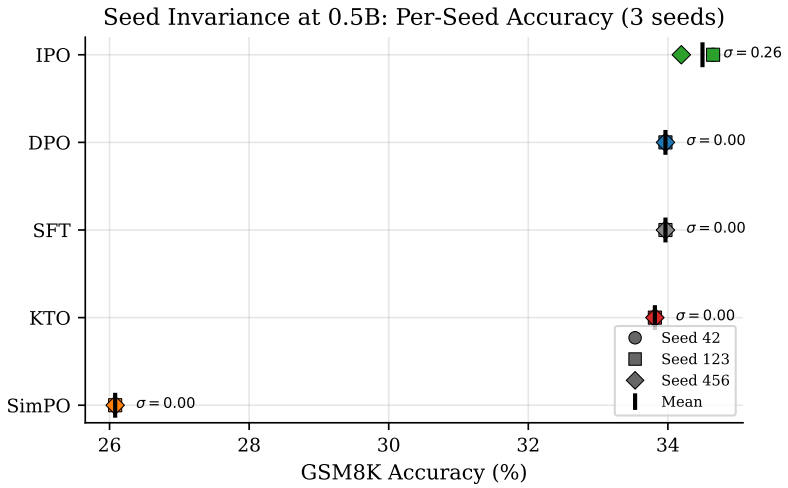


Figure 4: **0.5B is at the noise floor of our protocol, not a meaningful inter-algorithm comparison.** Per-seed GSM8K strict-match accuracy for each algorithm across 3 seeds. Five of six methods converge to identical accuracy ($\sigma$=0, all 33.81–33.97%); IPO is the only algorithm with non-trivial seed variance ($\sigma$=0.26). With greedy decoding and a 0.5B base that already scores 19.6% before training, post-training adds the same $\sim$14 pp lift regardless of loss. We retain 0.5B in the scaling plot for completeness, but *the 0.5B column should not be read as evidence that algorithms are equivalent*—it is evidence that 0.5B Qwen 2.5 Instruct on GSM8K is below the differentiation threshold of any of these methods. The scaling-study signal therefore begins at 1.5B.

Table 16: Strict-match vs. flexible-extract accuracy on GSM8K. At 3B, the format compliance gap ranges from $-26.2$ to $-33.7$ pp across trained models, confirming format non-compliance drives the low strict-match scores.

| Scale | Algorithm | Strict (%) | Flexible (%) | Gap (pp) |
|---|---|---|---|---|
| 1.5B | SP-RFT | 54.66 | 55.57 | $-0.9$ |
| | DPO | 51.18 | 49.81 | $+1.4$ |
| | IPO | 52.39 | 52.31 | $+0.1$ |
| | KTO | 51.71 | 51.18 | $+0.5$ |
| | SimPO | 43.44 | 43.06 | $+0.4$ |
| 3B (full FT) | Base | 10.69 | 62.93 | $-52.2$ |
| | SP-RFT | 18.04 | 44.28 | $-26.2$ |
| | DPO | 16.45 | 42.91 | $-26.5$ |
| | IPO | 16.83 | 45.72 | $-28.9$ |
| | KTO | 15.31 | 44.50 | $-29.2$ |
| | SimPO | 6.90 | 40.56 | $-33.7$ |
| 7B (LoRA) | SP-RFT | 76.42 | 80.29 | $-3.9$ |
| | DPO | 83.85 | 84.99 | $-1.1$ |
| | IPO | 81.35 | 83.09 | $-1.7$ |
| | KTO | 80.21 | 82.49 | $-2.3$ |
| | SimPO | 85.82 | 80.97 | $+4.9$ |

## L Sampling-Based Evaluation

We re-evaluate 1.5B and 7B checkpoints using nucleus sampling (top-$p = 0.9$, temperature $= 0.8$) with 8 samples per problem, reporting majority-vote accuracy. Core rankings are preserved: at 7B, DPO and SimPO remain at the top, SP-RFT at the bottom. The inter-algorithm spread compresses slightly, consistent with sampling partially compensating for format compliance differences.

Table 17: Compute budget breakdown on H100 80GB GPUs.

| Category | # Runs | Time/Run | GPUs | GPU-h |
|---|---|---|---|---|
| Core 0.5B SL (5 algs $\times$ 3 seeds) | 15 | 4–8 min | 1 | 1.5 |
| Core 0.5B RL (3 algs $\times$ 3 seeds) | 9 | $\sim$3 h | 2 | 54 |
| Core 1.5B SL (5 algs $\times$ 3 seeds) | 15 | 14–27 min | 1 | 5.0 |
| Core 3B SL full FT (3 algs) | 3 | 12–13 min | 1 | 0.6 |
| 3B LoRA factorial (3 algs) | 3 | 12–13 min | 1 | 0.6 |
| Core 7B SL (5 algs $\times$ 1 seed) | 5 | 335–460 min | 1 | 30 |
| Core 7B RL (SGRPO, 3 epochs) | 3 | $\sim$480 min | 2 | 48 |
| 7B multi-seed: SP-RFT/DPO/SimPO ($N$=5) | 13 | 330–360 min | 1 | 78 |
| 7B multi-seed: IPO/KTO ($N$=3) | 4 | 335–460 min | 1 | 27 |
| MATH 1.5B evals (5 methods) | 5 | $\sim$30 min | 1 | 2.5 |
| DPO variants 1.5B (20 $\times$ 5 seeds) | 100 | 14–27 min | 1 | 30 |
| DPO variants 7B (5 $\times$ 3 seeds) | 15 | 330–360 min | 1 | 90 |
| Ablations 1.5B | $\sim$57 | 9–10 min | 1 | 9 |
| Ablations 7B (LR sweep, KL, Base init) | 16 | 330–480 min | 1–2 | $\sim$100 |
| Gemma 3 (3 algs $\times$ 3 scales $\times$ 3 seeds) | 27 | varies | 1–2 | $\sim$75 |
| 7B Full FT factorial (SP-RFT/DPO/SimPO) | 3 | $\sim$180 min | 2 | 18 |
| 14B LoRA (SP-RFT/DPO/SimPO $\times$ 3 seeds) | 9 | $\sim$360 min | 2 | 108 |
| Fixed-dataset control (SP-RFT/DPO $\times$ 3 seeds) | 6 | 14–27 min | 1 | 2.0 |
| 1.5B LR sweep (SP-RFT/DPO $\times$ 4 LRs $\times$ 3 seeds) | 18 | 14–27 min | 1 | $\sim$5.0 |
| 3B LR sweep (SP-RFT/DPO $\times$ 3 LRs $\times$ 3 seeds) | 18 | $\sim$25 min | 1 | $\sim$7.5 |
| Gold-SFT control (1.5B $\times$ 3 + 7B $\times$ 3 seeds) | 6 | 27–360 min | 1 | $\sim$20 |
| **Total training** | $\sim$350 | – | – | **$\sim$720** |
| Evaluation (GSM8K, MATH, general) | $\sim$430+ | 5–600 min | 1 | $\sim$220 |
| **Grand total** | | | | **$\sim$940 GPU-h** |

Table 18: Multi-seed results at 1.5B on GSM8K (3 seeds each). Rankings preserved across seeds.

| Method | Original | Mean | $\sigma$ | s42 | s123 | s456 |
|---|---|---|---|---|---|---|
| SP-RFT | 54.66 | 54.36 | 0.59 | 53.90 | 55.19 | 53.98 |
| IPO | 52.39 | 52.24 | 0.22 | 52.54 | 52.16 | 52.01 |
| KTO | 51.71 | 51.15 | 1.77 | 51.33 | 53.22 | 48.90 |
| DPO | 51.18 | 49.08 | 0.61 | 49.58 | 49.43 | 48.22 |
| SimPO | 43.44 | 38.67 | 1.78 | 36.85 | 38.06 | 41.09 |

# M Learning Rate Sensitivity at 7B

Three observations emerge. **(1)** The default $10^{-6}$ is optimal for both SP-RFT and DPO; no higher LR improves either algorithm, ruling out under-training as an explanation for SP-RFT's 7B performance. **(2)** DPO is *more* LR-sensitive than SP-RFT: at $10^{-5}$, DPO collapses to $16.15\%$ (one seed hits $1.2\%$) while SP-RFT retains $53.12\%$. **(3)** The ranking inversion (DPO $>$ SP-RFT) holds at the independently optimal LR for each algorithm ($10^{-6}$ for both), confirming it is not an artifact of hyperparameter tuning.

**Note on the lower boundary.** Our grid does not extend below $10^{-6}$ (e.g., to $5{\times}10^{-7}$). Because performance degrades *monotonically and steeply* at every LR above $10^{-6}$—with DPO losing 27 pp at just $5\times$ higher—the gradient strongly favors the conservative end, and $10^{-6}$ already matches the recommended LoRA LR for both Qwen 2.5 and Gemma 3 in their official fine-tuning guides. Nonetheless, we cannot rule out a marginal improvement at $5{\times}10^{-7}$; if it exists, it would affect SP-RFT and DPO similarly and would not change the ranking inversion.

Table 19: Multi-seed results at 7B (LoRA) on GSM8K. Core algorithms have $N$=5 seeds; IPO/KTO have $N$=3.

| Method | $N$ | Mean | $\sigma$ | s42 | s456 | s789 | s1024 | s1337 |
|---|---|---|---|---|---|---|---|---|
| DPO | 5 | **83.38** | 0.56 | 83.85 | 82.79 | 83.02 | 84.23 | 83.02 |
| SimPO | 5 | 83.32 | 1.79 | 85.82 | 81.50 | 82.64 | 85.06 | 81.58 |
| IPO | 3 | 80.39 | 0.92 | 81.35 | — | 80.67 | — | 79.15 |
| KTO | 3 | 80.24 | 0.16 | 80.21 | — | 80.44 | — | 80.06 |
| SP-RFT | 5 | 77.38 | 1.11 | 76.42 | 79.30 | 76.65 | 76.57 | 77.94 |

Table 20: Learning rate sensitivity for SP-RFT and DPO at 7B (LoRA) on GSM8K (%). Each cell reports mean $\pm$ std over 3 seeds (SP-RFT) or 2 seeds (DPO). The default LR $= 10^{-6}$ is optimal for both algorithms; higher rates degrade performance monotonically.

| LR | SP-RFT Mean | SP-RFT Seeds | DPO Mean | DPO Seeds |
|---|---|---|---|---|
| $10^{-6}$ (default) | $77.38_{\pm 1.11}$ | 5 seeds | $83.38_{\pm 0.56}$ | 5 seeds |
| $5\times10^{-6}$ | $66.57_{\pm 5.14}$ | 66.3 / 60.4 / 73.0 | $56.75_{\pm 8.68}$ | 65.4 / 48.1 |
| $10^{-5}$ | $53.12_{\pm 4.90}$ | 59.3 / 47.3 / 52.8 | $16.15_{\pm 14.94}$ | 31.1 / 1.2 |
| $5\times10^{-5}$ | $30.20_{\pm 7.73}$ | 34.1 / 37.1 / 19.4 | — | — |
| $2\times10^{-4}$ | 0.23 | 1 seed | — | — |

## N Learning Rate Sensitivity at 1.5B

To directly address whether the 1.5B SP-RFT–DPO ranking is an artifact of the shared LR ($10^{-6}$), we sweep LR in $\{5\times10^{-7}, 10^{-6}, 5\times10^{-6}, 10^{-5}\}$ for both SP-RFT and DPO at 1.5B with 3 seeds each (24 runs total).

Table 21: Learning rate sensitivity for SP-RFT and DPO at 1.5B (full FT) on GSM8K (%). Each cell reports mean $\pm$ population std over 3 seeds. SP-RFT leads at *every* LR. DPO collapses catastrophically at $\geq 5\times10^{-6}$. At independently optimal LRs, SP-RFT@$10^{-5}$ (56.61%) beats DPO@$5\times10^{-7}$ (52.26%) by +4.35 pp.

| LR | SP-RFT Mean$_{\pm\sigma}$ | SP-RFT Seeds | DPO Mean$_{\pm\sigma}$ | DPO Seeds |
|---|---|---|---|---|
| $5\times10^{-7}$ | $53.83_{\pm 0.21}$ | 54.13 / 53.68 / 53.68 | $\mathbf{52.26}_{\pm 0.52}$ | 52.46 / 52.77 / 51.55 |
| $10^{-6}$ (default) | $54.36_{\pm 0.59}$ | 55.19 / 53.90 / 53.98 | $49.08_{\pm 0.61}$ | 49.43 / 49.58 / 48.22 |
| $5\times10^{-6}$ | $54.74_{\pm 1.21}$ | 56.25 / 53.30 / 54.66 | $16.00_{\pm 11.69}$ | 20.39 / 27.60 / 0.00$^{\dagger}$ |
| $10^{-5}$ | $\mathbf{56.61}_{\pm 0.48}$ | 55.95 / 56.79 / 57.09 | $19.66_{\pm 1.81}$ | 20.62 / 21.23 / 17.13 |

$^{\dagger}$Complete collapse: model generates only degenerate output.

Four observations: **(1)** SP-RFT wins at every LR tested. **(2)** At independently optimal LRs, the gap widens to +4.35 pp. **(3)** DPO is dramatically more LR-sensitive at 1.5B than at 7B: the usable LR range for DPO shrinks from roughly one order of magnitude at 7B to half an order at 1.5B. **(4)** The 1.5B and 7B sweeps tell opposite stories about optimal LR, confirming that LR sensitivity itself is scale-dependent.

## O Learning Rate Sensitivity at 3B

The 3B collapse (Section 4.1) is fully recoverable with a LR sweep. We sweep LR in $\{5\times10^{-7}, 5\times10^{-6}, 10^{-5}\}$ for both SP-RFT and DPO at 3B (full FT) with 3 seeds each (18 runs).

Three observations: **(1)** The 3B collapse is fully recoverable for SP-RFT: at $10^{-5}$, it reaches 55.70%—comparable to 1.5B Instruct (54.36%). **(2)** SP-RFT wins at every LR, preserving the small-scale ranking. **(3)** DPO is dramatically more LR-sensitive than SP-RFT at 3B, mirroring the pattern at 1.5B and 7B.

Table 22: Learning rate sensitivity for SP-RFT and DPO at 3B (full FT) on GSM8K (%). Each cell reports mean ± population std over 3 seeds. SP-RFT recovers to 55.7% at $10^{-5}$—comparable to 1.5B Instruct—confirming the default-LR collapse was a hyperparameter pathology. SP-RFT leads at every LR.

| LR | SP-RFT Mean$_{\pm\sigma}$ | SP-RFT Seeds | DPO Mean$_{\pm\sigma}$ | DPO Seeds |
|---|---|---|---|---|
| $5\times10^{-7}$ | $17.79_{\pm0.16}$ | 17.82 / 17.97 / 17.59 | $15.77_{\pm0.91}$ | 14.56 / 16.00 / 16.76 |
| $10^{-6}$ (default) | $18.35^{b}$ | — | $14.44_{\pm2.01}$ | 16.45 / 14.44 / 12.42 |
| $5\times10^{-6}$ | $41.80_{\pm3.84}$ | 40.26 / 38.06 / 47.08 | $\mathbf{34.55}_{\pm4.52}$ | 33.81 / 29.42 / 40.41 |
| $10^{-5}$ | $\mathbf{55.70}_{\pm1.09}$ | 54.21 / 56.79 / 56.10 | $16.30_{\pm7.34}$ | 5.91 / 21.46 / 21.53 |

[b] Single seed from Table 1.

Table 23: SGRPO at 7B with and without KL regularization on GSM8K (%).

| Method | $\lambda_{\text{KL}}$ | GSM8K (%) | Δ vs. DPO |
|---|---|---|---|
| SGRPO (default) | 0 | 80.59 | −2.79 |
| SGRPO (regularized) | 0.01 | 80.74 | −2.64 |
| DPO (reference) | $\beta = 0.1$ | $83.38_{\pm0.56}$ | — |

# P KL Regularization Ablation for SGRPO

Adding KL regularization yields only +0.15 pp, confirming the online–offline gap is genuine rather than a regularization artifact.

# Q Base vs. Instruct Initialization at 7B

Table 24: Base vs. Instruct initialization at 7B on GSM8K (%). The spread collapses from 6.00 pp to 1.82 pp. SP-RFT leads from Base, reversing the Instruct ranking.

| Algorithm | From Base (7B) | From Instruct (7B) | Δ |
|---|---|---|---|
| SP-RFT | **82.49** | $77.38_{\pm1.11}$ | +5.11 |
| DPO | 80.67 | $83.38_{\pm0.56}$ | −2.71 |
| SimPO | 80.82 | $83.32_{\pm1.79}$ | −2.50 |
| *Spread* | *1.82 pp* | *6.00 pp* | |

From Base initialization at 7B, the ranking (SP-RFT > SimPO > DPO) differs from both the 1.5B Base ranking (SimPO > DPO > SP-RFT) and the 7B Instruct ranking (DPO > SimPO ≫ SP-RFT). A residual ranking shift between scales persists even from Base, but at ≤1.82 pp—far smaller than the 6.00 pp Instruct spread. The key finding is that initialization accounts for the majority of inter-algorithm variance (3.3× compression at 7B, 14.7× at 1.5B), even though a small scale-dependent residual remains.

# R Hyperparameter Sensitivity

# S 3B Flexible Extraction Analysis

Under flexible extraction, the 3B base model achieves 62.93%—comparable to the 1.5B Instruct model's strict-match performance. This reveals that the Qwen 2.5 3B Instruct model can *solve* GSM8K problems at a competitive rate but fails to produce answers in the expected format. Post-training partially addresses this: SP-RFT reduces the format gap from −52.2 pp (base) to −26.2 pp.

Table 25: $\beta$ sensitivity at 1.5B on GSM8K (%). Both variants are stable across a $50\times$ range; default $\beta = 0.1$ is near-optimal.

| **Variant** | $\beta = 0.01$ | $\beta = 0.1$ | $\beta = 0.5$ | **Range** |
|---|---|---|---|---|
| DPOP | 54.66 | **55.19** | 53.30 | 1.89 pp |
| ORPO | 54.81 | 54.44 | 53.83 | 0.98 pp |
| *Vanilla DPO* | *49.76*$_{\pm 2.27}$ *(variant sweep)* | | | |

# T General Benchmark Results

Table 26 reports ARC-Challenge, HellaSwag, and WinoGrande accuracy across scales and architectures. At every scale and architecture, the spread is $<2$ pp—confirming that post-training algorithms do not meaningfully differ on these benchmarks. **This is a critical scope qualifier for our main findings:** the ranking inversion documented in this paper is specific to mathematical reasoning (GSM8K, MATH). On these general benchmarks, no inversion is observable, and algorithm choice is essentially irrelevant.

**Caveat: benchmark format.** ARC-C, HellaSwag, and WinoGrande are multiple-choice or perplexity-based evaluations, not open-ended generative tasks. Since our analysis identifies format compliance—a generative property—as the key differentiator at small scales (Section 4.4), the absence of algorithmic differences on non-generative benchmarks is expected and does not rule out the possibility that inversions occur on other *generative* tasks. We leave this to future work.

Table 26: General benchmark accuracy (%) after post-training. Spread remains $<2$ pp across all scales and architectures, confirming that algorithmic differences are task-specific.

| **Architecture** | **Algorithm** | **ARC-C** | **HellaSwag** | **WinoGrande** | **Avg** |
|---|---|---|---|---|---|
| Gemma 1B | Base | 38.7 | 57.7 | 58.9 | 51.8 |
| | SP-RFT | 39.1 | 57.9 | 59.1 | 52.0 |
| | DPO | 38.7 | 57.6 | 58.8 | 51.7 |
| | SimPO | 39.6 | 57.2 | 57.6 | 51.5 |
| Gemma 12B | Base | 61.1 | 81.9 | 74.6 | 72.5 |
| | SP-RFT | 60.1 | 81.7 | 74.5 | 72.1 |
| | DPO | 58.4 | 81.2 | 74.6 | 71.4 |
| | SimPO | 56.7 | 81.3 | 73.6 | 70.5 |
| Qwen 7B FT | Base | 55.2 | 80.5 | 71.1 | 68.9 |
| | SP-RFT | 56.1 | 80.6 | 71.0 | 69.2 |
| | DPO | 55.0 | 80.6 | 70.4 | 68.7 |
| | SimPO | 54.1 | 80.6 | 69.9 | 68.2 |
| Qwen 14B | Base | 62.4 | 84.4 | 76.0 | 74.3 |
| | SP-RFT | 61.9 | 84.5 | 75.2 | 73.9 |
| | DPO | 61.8 | 84.4 | 75.4 | 73.9 |
| | SimPO | 61.7 | 84.5 | 74.8 | 73.7 |